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Report of Findings

**Belle Realty of Lafayette
Claim No: 23595506**

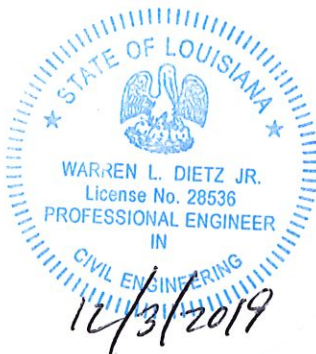
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Prepared For:

**Liberty Mutual Insurance Company
7900 Windrose Avenue
Plano, Texas 75024**

Attention:

Mr. Jeff DeBenedetto




**Warren L. Dietz Jr., P.E.
Louisiana Engineer Number 28536
Senior Consultant**

TABLE OF CONTENTS

I.	Introduction	1
II.	Conclusion	2
III.	Discussion	3
	• General History	
	• Observations	
	• Weather Information	
	• Analysis	
IV.	Basis of Report	12
V.	Attachments	13
	A. Photographs	
	B. Curriculum Vitae	

Section I

INTRODUCTION

On October 31, 2019, representatives of Belle Realty of Lafayette reported that hail impacts damaged the three-tab asphalt-composition shingles and gravel ballasted built-up system on the various roof slopes over their building. The buildings, which was a strip mall, were located at 101-149 Arnould Boulevard in Lafayette, Louisiana.

Rimkus Consulting Group, Inc. (Rimkus) was retained by Liberty Mutual Insurance Company to determine if the roof shingles and modified bitumen had been damaged from hail impacts. Warren L. Dietz Jr., P.E., performed our inspection and prepared this report. This **Report of Findings** was reviewed by Tony R. Clark, P.E., Principal Consultant.

This report was prepared for the exclusive use of Liberty Mutual Insurance Company and is not intended for any other purpose. Our report was based on information made available to us at the time, as described in **Basis of Report**. Should additional information become available, we reserve the right to determine the impact, if any, the new information may have on our opinions and conclusions and to revise our opinions and conclusions if necessary and warranted.

Section II

CONCLUSIONS

1. There were numerous small, rounded areas of granule loss consistent with hail impacts, on the three-tab asphalt-composition shingles on the surfaces of the west-facing roof slopes over the building. The hail damage appeared old and prior to the reported 2019 storm event.
2. There was no physical evidence to suggest that the three-tab asphalt-composition shingles on the north-facing, south-facing, and east-facing roof slopes over the building were damaged from hail impacts.
3. There was no physical evidence to suggest that the thermoplastic polyolefin (TPO) membrane on a portion of the low-sloped roof over the building was damaged from hail impacts.
4. There was no physical evidence to suggest that the gravel ballasted built-up roof system was damaged from hail impacts.

Section III

DISCUSSION

General History

The Belle Realty of Lafayette building, which was built in phases between 1977 and 2004, was a one-story, steel-framed, commercial structure that was supported by a concrete slab-on-ground foundation system. The roof surfaces were covered with three-tab asphalt-composition shingles, gravel ballasted built-up system, and TPO membrane. The exterior claddings were brick veneer, stucco, and wood siding. A published map showed the front of the building faced east-northeast. For discussion purposes, the front was referenced to face east (**Photograph 1**).

During the course of our site visit on November 5, 2019, Mr. Adam Abdalla, representative of Belle Realty of Lafayette, reported the following regarding the buildings:

- A hailstorm in the beginning of 2019 damaged the coverings on the surfaces of the various roof slopes over the building.
- There are now dark stains in the suspended acoustical tiles on the ceilings above Belle Realty, Jason's Deli, Franklin Financial, Politics, Lola Caroline Company, Great American Cookie's, and Brothers. These are all businesses leasing space in the building.

Observations

During the course of our site visit on November 5, 2019, we observed the following regarding the exterior of the building:

- There were no missing or detached sections of wood siding on the various sides of the building.

- There were no spatter marks or small rounded areas of displaced wood fibers on the wood siding on the various sides of the building.
- There were no broken glass panes in any of the windows on the various sides of the building.
- There were no indentations in the light-gauge metal gutters and downspouts on the various sides of the building (**Photograph 2**).
- There were no spatter marks or indentations on the heavy-gauge electrical boxes, which were located on the west side of the building (**Photograph 3**).
- There were no spatter marks or small rounded areas of displaced wood fibers on the wood fencing at the site (**Photograph 4**).

During the course of our site visit on November 5, 2019, we observed the following regarding the various roof surfaces covered with three-tab asphalt-composition shingles:

North-Facing Roof Slope

- There were four roof slopes that faced toward the north.
- There were no missing or torn three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- There were slipped shingle tabs on one of these roof slopes (**Photograph 5**).
- There was evidence, mastic, of prior repairs to the shingle tabs on one of these roof slopes (**Photograph 6**).
- The tar seals for the three-tab asphalt-composition shingles on the surfaces of these roof slopes were not easily broken with hand pressure. You could not break the tar seal without tearing the shingle.

- There was minor granule loss from the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- There was minor pitting and deterioration in the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- A detailed visual inspection of one test square (100 square feet) of roofing on these roof slopes revealed spatter marks but no small, rounded areas of granule loss consistent with hail impacts on the three-tab asphalt-composition shingles on the surfaces of these roof slopes (**Photograph 7**).

South-Facing Roof Slopes

- There were two roof slopes that faced toward the south.
- There were missing or torn three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- The tar seals for the three-tab asphalt-composition shingles on the surfaces of these roof slopes were not easily broken with hand pressure. You could not break the tar seal without tearing the shingle.
- There was minor granule loss from the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- There was minor to moderate pitting and deterioration in the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- A detailed visual inspection of one test square (100 square feet) of roofing on these roof slopes revealed spatter marks but no small, rounded areas of granule loss consistent with hail impacts on the three-tab asphalt-composition shingles on the surfaces of these roof slopes (**Photograph 8**).

East-Facing Roof Slopes

- There were three roof slopes that faced toward the east.
- There was no missing, torn, creased, or slipped three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- The tar seals for the three-tab asphalt-composition shingles on the surfaces of these roof slopes were not easily broken with hand pressure. You could not break the tar seal without tearing the shingle.
- There was minor granule loss from the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- There was minor to moderate pitting and deterioration in the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- A detailed visual inspection of one test square (100 square feet) of roofing on these roof slopes revealed spatter marks but no small, rounded areas of granule loss consistent with hail impacts on the three-tab asphalt-composition shingles on the surfaces of these roof slopes (**Photograph 9**).

West-Facing Roof Slopes

- There were three roof slopes that faced toward the west.
- There were isolated locations where there were missing and torn three-tab asphalt-composition shingles on the surfaces of these roof slopes (**Photograph 10**).
- There were isolated locations where there was evidence, different-colored shingles, of prior repairs to the three-tab asphalt-composition shingles on the surfaces of these roof slopes (**Photograph 11**).

- The tar seals for the three-tab asphalt-composition shingles on the surfaces of these roof slopes were not easily broken with hand pressure. You could not break the tar seal without tearing the shingle.
- There was minor to substantial granule loss from the three-tab asphalt-composition shingles on the surfaces of these roof slopes. In some instances, the fibers of the shingles were exposed (**Photograph 12**).
- There was minor to moderate pitting and deterioration in the three-tab asphalt-composition shingles on the surfaces of these roof slopes.
- A detailed visual inspection of one test square (100 square feet) of roofing on these roof slopes revealed numerous spatter marks and small, rounded areas of granule loss consistent with hail impacts on the three-tab asphalt-composition shingles on the surfaces of these roof slopes (**Photograph 13**).
- There were indentations in the light-gauge metal roof vents within these roof slopes. The indentations were estimated to be pea-sized in diameter (**Photograph 14**).

During the course of our site visit on November 5, 2019, we observed the following regarding the various low-sloped roof surfaces over the building:

- The majority of the low-sloped roof over the building consisted of a gravel ballasted built-up system (**Photograph 15**). There were no crushed or displaced sections of gravel ballast on the built-up system over the building.
- There were no visible punctures in the gravel ballasted built-up system on the low-sloped roof surface over the building.
- There was a TPO membrane on a portion of the low-sloped roof over the building.

- There were no punctures in the TPO membrane on the low-sloped roof over the building (**Photograph 16**).
- There were indentations in the cooling fins of the air-conditioning compressors on the surface of the low-sloped roof over the building (**Photographs 17**).
- There were indentations in the light-gauge metal turbines within the low-sloped roof over the building. The indentations were estimated to be pea-sized in diameter (**Photograph 18**).

Weather Information

A Hail Verification Report (Report 8137944) from CoreLogic indicated that there was one recorded hail event with maximum-sized hailstones of 1.1 inches at the location of the Belle Realty of Lafayette building between January 1, 2009, and November 12, 2019. This event occurred on April 2, 2017. There were two recorded hail events with maximum-sized hailstones of 0.75 inch and 1.1 inches within one mile of the Belle Realty of Lafayette building between January 1, 2009, and November 12, 2019. These events occurred on March 18, 2016 and April 2, 2017. The hail size was determined by radar signature and was not field verified.

Analysis

There were numerous small, rounded areas of granule loss consistent with hail impacts, on the three-tab asphalt-composition shingles on the surfaces of the west-facing roof slopes over the building. The hail damage appeared old and prior to the reported 2019 storm event. Composition shingle roofs damaged by hail are characterized by random-sized bruises to the shingles distributed fairly evenly over the areas of all the roof slopes exposed to the hail. When it occurs, hail damage will be found throughout areas of the roof system that are exposed to hail, and will not occur in grouped clusters. Damage will be found on the edges of the shingles, as well as in the fields of the mats. Shingles that make up the roof ridges, or unsupported shingles, such as those at ridge vents and roof edges are particularly vulnerable to hail damage. A detailed visual inspection of

one test square of roofing (100 square feet), which was made on all of the various roof slopes over the building, revealed spatter marks and small, rounded areas of granule loss consistent with hail impacts on the west-facing roof slopes over the building. A Hail Verification Report (Report 8137944) from CoreLogic indicated that there was one recorded hail event with maximum-sized hailstones of 1.1 inches at the location of the Belle Realty of Lafayette building between January 1, 2009, and November 12, 2019. This event occurred on April 2, 2017. There were two recorded hail events with maximum-sized hailstones of 0.75 inch and 1.1 inches within one mile of the Belle Realty of Lafayette building between January 1, 2009, and November 12, 2019. These events occurred on March 18, 2016 and April 2, 2017. The hail size was determined by radar signature and was not field verified. This was consistent with the observations, which as mentioned above, indicated that there were spatter marks and small, rounded areas of granule loss consistent with hail impacts on the west-facing roof slopes over the building.

In addition, other objects, such as gutters, downspouts, metal vent caps, and air-conditioner coil fins, will also be struck by hail and will exhibit evidence of such in the form of indentations or spatter marks. In most instances, the impact energy required to damage these objects is less than that required to damage typical roof shingles. This is due to the lack of rigid support to these objects and their ductility. Consequently, indentations in these objects from hailstones do not necessarily indicate damage to the shingles. However, indentations may be used to evaluate the size of the hail that struck a particular roof system. Our observations revealed that there were no indentations in the light-gauge metal gutters and downspouts on the various sides of the building. There were no spatter marks or indentations on the heavy-gauge electrical boxes, which were located on the west side of the building. There were no spatter marks or small rounded areas of displaced wood fibers on the wood fencing at the site. There were indentations in the light-gauge metal roof vents within the west-facing roof slopes over the building. The indentations were estimated to be pea-sized in diameter. There were indentations in the cooling fins of the air-conditioning compressors on the surface of the low-sloped roof over the building. There were indentations in the light-gauge

metal turbines within the low-sloped roof over the building. The indentations were estimated to be pea-sized in diameter.

According to Hail-Fall, Roofing, and Impact Testing, damage to roofing is defined as “a diminution of water-shedding capability or a reduction in the expected long-term service life of the roofing material.” For composition shingles, damage is defined as punctures, tears, or bruises in the shingle mat or displacement of granules to the extent that the bitumen becomes exposed.

There was no physical evidence to suggest that the three-tab asphalt-composition shingles on the north-facing, south-facing, and east-facing roof slopes over the building were damaged from hail impacts. Composition shingle roofs damaged by hail are characterized by random-sized bruises to the shingles distributed fairly evenly over the areas of all the roof slopes exposed to the hail. When it occurs, hail damage will be found throughout areas of the roof system that are exposed to hail, and will not occur in grouped clusters. Damage will be found on the edges of the shingles, as well as in the fields of the mats. Shingles that make up the roof ridges, or unsupported shingles, such as those at ridge vents and roof edges are particularly vulnerable to hail damage. A detailed visual inspection of one test square of roofing (100 square feet), which was made on all of the north-facing, south-facing, and east-facing roof slopes over the building, revealed spatter marks on the north-facing roof slopes, but no small, rounded areas of granule loss consistent with hail impacts on the north-facing, south-facing, or east-facing roof slopes over the building.

There was no physical evidence to suggest that the TPO membrane on a portion of the low-sloped roof over the building was damaged from hail impacts. From viewing the roof surface over the building, our observations indicated that there was a TPO membrane on a portion of the low-sloped roof surface over the building. There were no punctures in the TPO membrane on the low-sloped roof over the building. This supports our conclusion that there was no physical evidence to suggest that the TPO membrane on a portion of the low-sloped roof over the building was damaged from hail impacts.

There was no physical evidence to suggest that the gravel ballasted built-up roof system was damaged from hail impacts. From viewing the roof surface over the building, our observations indicated that there were no crushed or displaced sections of gravel ballast on the built-up system over the building. There were no visible punctures in the gravel ballasted built-up system on the low-sloped roof surface over the building. This supports our conclusion that there was no physical evidence to suggest that the gravel ballasted built-up roof system was damaged from hail impacts.

Section IV

BASIS OF REPORT

1. Warren L. Dietz Jr., P.E., interviewed Mr. Adam Abdalla, representative of Belle Realty of Lafayette, on November 5, 2019.
2. Warren L. Dietz Jr., P.E., inspected the Belle Realty of Lafayette building on November 5, 2019.
3. Photographs were taken of the Belle Realty of Lafayette building by Warren L. Dietz Jr., P.E., on November 5, 2019.
4. We reviewed a Hail Verification Report (Report 8137944) from CoreLogic from January 1, 2009, through November 12, 2019, at the location of the Belle Realty of Lafayette building.
5. *Hail-Fall, Roofing, and Impact Testing*, HAAG Engineering, 2004 was reviewed.

Section V
ATTACHMENTS

A. Photographs

B. Curriculum Vitae

Section V
ATTACHMENT A

Photographs

Photographs taken during our inspection, which were not included in this report, were retained in our files and are available to you upon request.

Photograph 1

View of the east (front) elevation of the Belle Realty of Lafayette building.



Photograph 2

View of a light-gauge metal gutter on the east side of the building. Notice no spatter marks or indentations in the gutter.



Photograph 3

View of an electrical panel on the west side of the building. Notice no spatter marks or indentations on the surface of the electrical panel.



Photograph 4

View of a section of wood fencing, which faced toward the north, on the west side of the building. Notice no small rounded areas of displaced wood fibers on the fencing.



Photograph 5

View of the slipped shingles on one of the north-facing roof slopes over the building.



Photograph 6

View of evidence, mastic, of prior repairs to the shingle tabs on one of the north-facing roof slopes over the building.



Photograph 7

View of the shingles on the surface of a north-facing roof slope over the building. Notice no spatter marks or small, rounded area of granule loss on the shingles.



Photograph 8

View of the shingles on the surface of a south-facing roof slope over the building. Notice no spatter marks or small, rounded area of granule loss on the shingles.



Photograph 9

View of the shingles on the surface of an east-facing roof slope over the building. Notice no spatter marks or small, rounded area of granule loss on the shingles.



Photograph 10

View of the torn and missing shingle tabs on one of the west-facing roof slopes over the building.



Photograph 11

View of the evidence, different-colored shingles, of prior repairs to the three-tab asphalt-composition shingles on a west-facing roof slope over the building.



Photograph 12

View of the granule loss from the shingle tabs on one of the west-facing roof slopes over the building. Notice the fibers in the shingles are exposed.



Photograph 13

View of the shingles on the surface of the west-facing roof slope over the building. Notice the spatter marks and small, rounded areas of granule loss on the shingles.



Photograph 14

View of the pea-sized indentations in a light-gauge metal vent within a west-facing roof slope over the building.



Photograph 15

View of the built-up system with a gravel ballast on the low-sloped roof over the building.



Photograph 16

View of the TPO membrane on a portion of the low-sloped roof over the building. Notice no punctures in the TPO membrane.



Photograph 17

View of the indentations in the cooling fins of the air-conditioning compressors on the surface of the low-sloped roof over the building.



Photograph 18

View of the pea-sized indentations in a light-gauge metal turbine within the low-sloped roof surface over the building.



Section V
ATTACHMENT B

Curriculum Vitae



Warren L. Dietz, P.E.

Senior Consultant
Property Division

Background

Mr. Dietz holds a B.S. degree in Civil Engineering and is a registered professional engineer in Louisiana. He has over 18+ years of experience as a project manager and engineer developed with various engineering firms.

Mr. Dietz has “hands on” experience in the design and construction management of residential buildings and most general civil projects. The design experience varies from conceptual and planning, to final plans and specifications. Project types have included roadways, foundations for residential and commercial buildings for private entities. Mr. Dietz has been involved with and associated with most aspects of residential construction, including foundation design, structural plan review, structure and foundation damage inspection related to storms and soil issues.

As a forensic engineer, Mr. Dietz has investigated wind damage to residential and commercial roofs, differential foundation movement, moisture source determinations, potable water and sanitary sewer line failures, wind damage to residential and commercial structures, storm surge damage to residential and commercial structures, and fire related structural damage to residential and commercial structures. He has performed structural condition assessments for structures damaged by automobile impacts, tree impacts, tornados, and vibrations from construction operations. He has also performed causation analysis for collapses of residential structures and roof collapses from overloading and faulty design. He has also performed structural condition analysis for damages to in-ground swimming pools. He has performed slip and fall analysis and trip and fall analysis for residential and commercial buildings. This includes analyzing ramps, staircases, walkways, and floor surfaces in residential and commercial buildings. Finally, he has assessed lighting conditions for commercial buildings and outdoor areas involved in premises liability claims.

Contact Information

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Professional Engagements

- **Highways**

- State Highway 23 – Plaquemines Parish, LA. Performed engineering calculations for quantities of asphaltic concrete, aggregate surface course material, cold planning, cement stabilization and guard rail. Prepared cost estimates for the construction of each overlay project.

- **Residential/Commercial**

- Custom homes – Lakeshore Estates Subdivision, Slidell, LA (2004 and 2005). Responsible for foundation design, structural review of architectural plans, and overseeing the daily construction activities.

Forensic Engagements

- **Building Envelope**

- Investigated differential foundation movement.
- Moisture source determination for roof leaks, roof flashing defects, oversized HVAC units, window flashing defects, stucco defects, and EIFS defects.
- Structural condition assessments for residential and commercial buildings damaged from storm surge and wind forces associated with hurricanes.
- Structural condition assessments for structures damaged by tornados.
- structural condition assessments for structures damaged by automobile impacts.
- Performed causation analysis for collapses of residential and commercial structures from overloading, faulty design, and installation defects.
- Performed structural condition analysis for damages to in-ground swimming pools.
- Performed slip and fall and trip and fall investigations.
- Assessment of damage to ceramic tile.
- Assessment of damage to wood planks from ground water migration and inadequate crawlspace ventilation.
- Assessment of damage to concrete slabs and driveways from soil heave, differential foundation movement, and stresses associated with drying and shrinkage.
- Assessment of damage to bulkheads from installation defects.

- **Roof Systems**

- Hail damage evaluation to three-tab asphalt composition shingles, laminated asphalt composition shingles, ribbed metal panels, standing seam metal panels, modified bitumen, and rubber membrane roof coverings on residential and commercial buildings throughout Louisiana, Mississippi, and Florida.

- **Disaster Investigations**

- Hurricane Michael – Florida (2018 and 2019)
- Hurricane Irma – Florida (2018)

- Hurricane Harvey – Louisiana (2017)
- Hurricane Isaac – Louisiana (2012 and 2013)
- Hurricane Gustav – Louisiana (2008 and 2009)
- Hurricane Katrina – Louisiana (2005, 2006, and 2007)

Professional Experience

- **Rimkus Consulting Group, Inc.** **2005 – Present**
 - Senior Consultant – Property Division
Responsible for investigating and evaluating commercial and residential structures to determine the cause of apparent structural concerns or construction defects.
- **C&C Construction, Inc.** **2004 – 2005**
 - Project Engineer
Responsible for the design and construction of custom residential homes in Jefferson, Orleans, and St. Tammany parishes. Reviewed architectural plans for structural soundness for multiple custom homes. Designed foundations for multiple custom homes. Oversaw the daily construction operations for multiple custom homes. Preparing cost estimates for the construction of residential buildings
- **Metal Pro Industries** **2003 – 2004**
 - Design Engineer
Responsible for the design of light gauge steel residential and commercial structures. Designed (size) wall studs, ceiling joists and roof rafters used in the light gauge steel construction of residential and commercial buildings. Designed foundations for residential and commercial buildings. Inspected construction sites and made sure building conforms to state building code. Reviewed architectural plans from structural soundness.
- **Digital Engineering and Imaging** **2002 – 2003**
 - Design Engineer
Responsible for the design of roadway overlay projects in Jefferson, parishes. Reviewed surveys prior to the design of roadway overlays to locate traffic loops, guard rails, turn lanes, cross-drains, box culverts, ditches, driveways, turnouts, RR crossings, catch basins, manholes and other field data necessary to design an overlay project. Performed engineering calculations for quantities of asphaltic concrete, aggregate surface course material, cold planning, cement stabilization and guard rail. Prepared cost estimates for the construction of each overlay project.
- **Louisiana Dept. of Transportation and Development** **1993 – 2002**
 - Design Engineer
Responsible for the design of overlay projects in District 02, which includes Jefferson, Lafourche,

Orleans, Plaquemines, St. Bernard, St. Charles and Terrebonne Parishes. Worked with the survey crew in the field cross-sectioning roadway, locating traffic loops, guard rails, turn lanes, cross-drains, box culverts, ditches, driveways, turnouts, RR crossings, catch basins, manholes and other field data necessary to design an overlay project. Performed engineering calculations for quantities of asphaltic concrete, aggregate surface course material, cold planning, cement stabilization and guard rail. Prepared cost estimates for the construction of each overlay project.

Education and Certifications

- **Civil Engineering, B.S.:** Louisiana State University (1991)
- **Registered Professional Engineer:** Louisiana, Mississippi, and Florida

Continuing Education

- Roofing Industry Committee on Weather Issues – RICOWI (Sept. 2007)
- Wind Loads for Buildings and Other Structures Seminar – ASCE (Jan. 2009)
- Premises Liability Seminar – Rimkus (Oct. 2009)
- Deposition and Trial Testimony Seminar – Rimkus (Feb. 2010)